MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION, March 2026

FOURTH SEMESTER

DSE: MG4DSEBTS201

RESEARCH METHODOLOGY AND BIOSTATISTICS

PART A

Multiple choice questions

Answer all questions

Each carries 1 mark

1) A population census that systematically describes demographic characteristics is an example of:
   Ans:Option c

2) Grouping people as male and female is an example of:
   Ans: Option c

3) ANOVA test statistic is:

   Ans: Option c

4) The discussion section of a research report is used to:
   Ans: Option d

5) Which of the following is an example of a limitation in research?
   Ans: Option c

6) An alternative hypothesis (H1) indicates that:
   Ans: Option c

7) Which method of data collection covers the widest area:
   Ans: Option a

8) Which one is a random sampling?
   Ans: Option a

9) A project proposal is mainly prepared to:
   Ans: Option a

10) In which method does the investigator fill the answers?
Ans: Option b

PART B
Answer any 10 questions
Each question carries 3 marks

11) Define bibliography and explain its purpose.
Definition of bibliography (1 mark)
Explanation (2 marks)

12) Explain the role of citation indexes in academic research and publication.
Definition of Citation Index (1 mark)
Purpose and uses (1 mark)
Major Citation Indexes (any two 0.5 marks each)

13) Define Sample space.
Definition and example of Sample space (1 mark). Explanation of discrete and continuous sample space with an example (2 marks)

14) What is budget justification in a research proposal?
Definition of budget justification (1 mark)
Its role in research proposal (2 marks)

15) Explain paired t test.
Brief explanation of Paired t test (1 mark). Objective of test (1 mark) . Formula and criteria (1 mark).

16) List out the steps to define a research problem
Define a research problem (1 mark)
List out the steps to set the research problem (minimum 4 points, 0.5 marks each)

17) What are the main characteristics of research?
Write any six characteristics of research (0.5 marks for each)

18) Explain the difference between results and discussion sections in a journal paper.
Explain the two sessions (1.5 marks)
Compare and write the major differences (1.5 marks)

19) Briefly explain simple bar diagram

Explanation of Simple bar diagram and its characteristics (2 marks). An example of simple bar diagram (1 mark).

20) What is secondary data? Give two examples of secondary data.
Definition of secondary data (1 mark)
Write any two examples with a brief explanation (2 marks)

21) What is a conference? Explain its main purpose.
Definition of conference (1 mark)
Comment on its purpose (2 marks)

22) Briefly explain SPSS.
Full form of SPSS and brief explanation (1 mark) .Applications (2 marks).

PART C

Answer any 6 questions

Each question carries 5 marks

23) Write an essay on a research paper and its significance in research communication.
What is a research paper (1 marks)
Briefly explain its steps (2 marks)
Significance in research communication (2 marks)

24) Write a short essay on interpretation with emphasise to its techniques and precautions to be taken.
What is interpretation? (1 mark)
Why is interpretation? (1 mark)
Techniques of interpretation (1.5 marks)
Precautions in interpretation (1.5 marks)

25) Distribution of MBBS students according to blood group is as follows. Represents the data using simple bar diagram.
Brief explanation of simple bar diagram (2marks). Presentation of data (3 marks)

26) Explain the need of a good research design and its features to execute a proper research.

What is a research design(1 marks)
Need of a good research design (2 marks)
Features of a good research design (2 marks)

27) What are the principles of good editing ?
Definition and brief explanation (2 marks). Objectives, Guidelines for good data editing and advantages (3 marks)

28) Explain steps in hypothesis testing.
Write the 5 steps for hypothesis testing : 1 mark for each steps.

29) Distinguish between discrete and continuous frequency table.
Format of a frequency table with relevant points (2 marks) and explanation using examples of discrete and continuous frequency tables (3 marks)

30) Write a short essay on various research reports.
Definition of research report. (1 mark)
Two types of reports and its explanation (2 marks each)